

Fitra Application

Smart filtering. Safe videos. Happy kids.



Computer Science

Overview

Fitra is an AI-powered Android application designed to create a safe video streaming environment for children. In today's digital world, many parents are concerned about the exposure of their kids to inappropriate content, especially LGBTQ+ themes that conflict with cultural and family values.

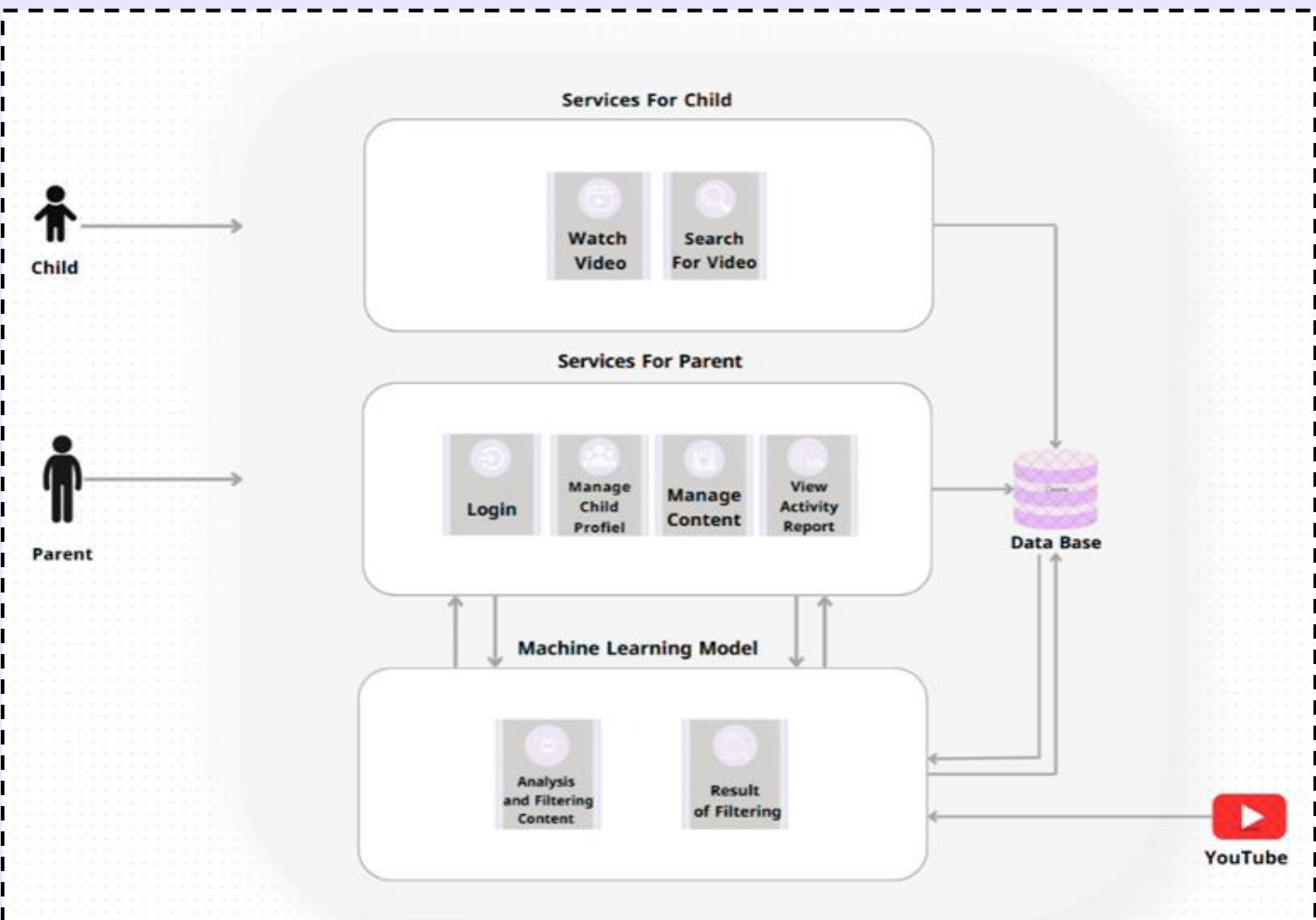
Fitra addresses this by analyzing YouTube videos using machine learning and natural language processing to detect and filter such content. The platform allows parents to create multiple child accounts, monitor watch and search history, and receive detailed activity reports. With an intuitive interface and advanced filtering system, Fitra offers a family-first solution that protects children while still allowing them to learn and enjoy content safely.

Objectives

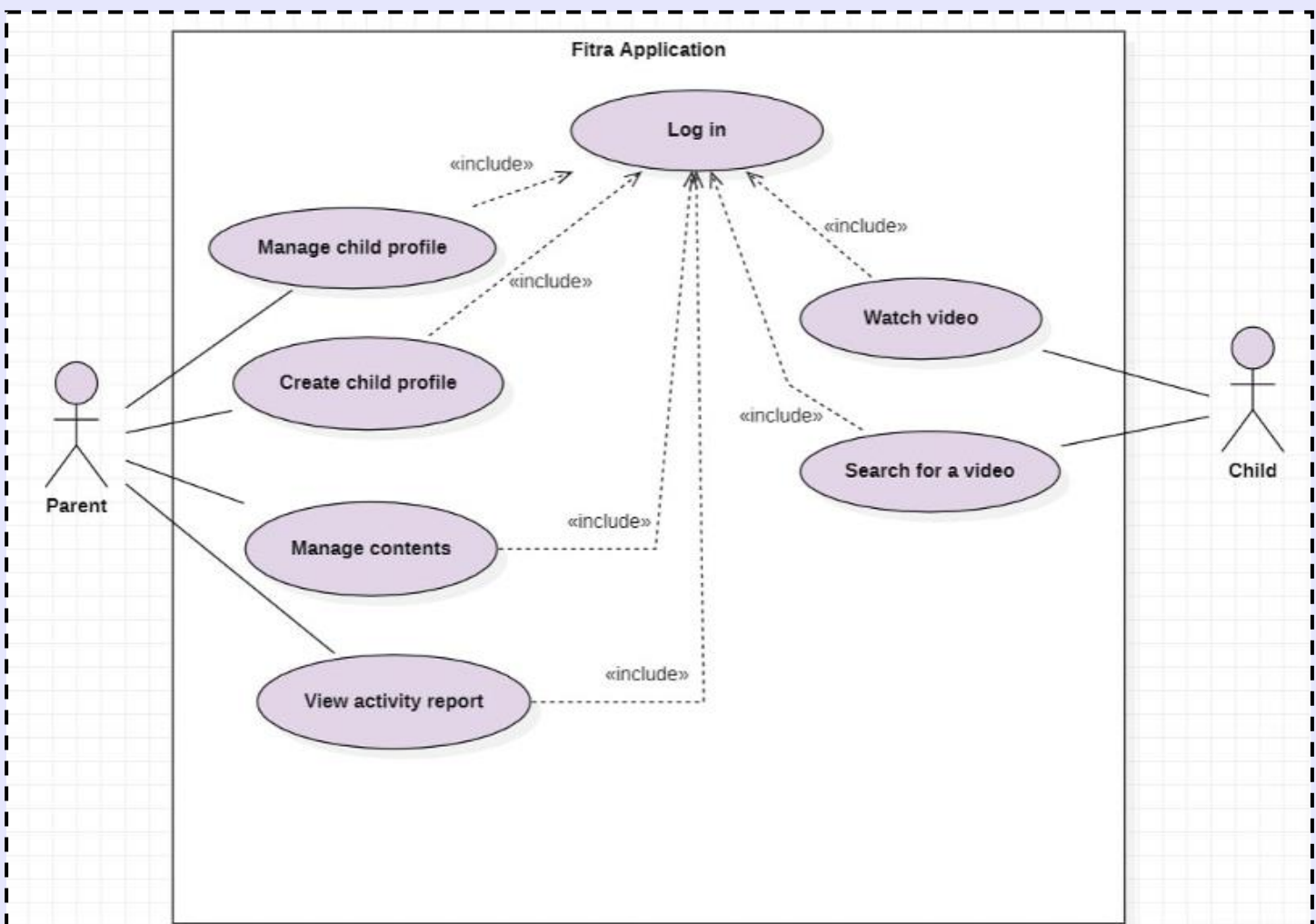
- It has user-friendly interfaces that children and parents can use effectively.
- Uses a machine-learning model to filter out LGBTQ+ content efficiently.
- Provide filtered content from YouTube that is free of LGBTQ+ content.
- Provide strong parental controlling tools that help parents control the contents their children watch and monitor all their activities on the application.

System Design

Architectural design



Use case

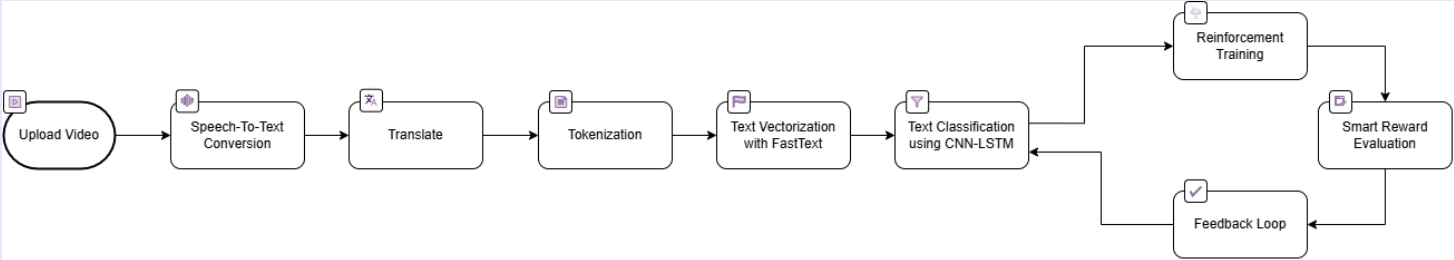


AI-Powered Filtering

Preprocessing

To prepare video content for classification, we developed a multi-step preprocessing that transforms raw YouTube data into clean, meaningful input for the machine learning model.

1. Audio Extraction
Downloaded audio from YouTube videos using yt-dlp to ensure full speech access regardless of subtitle availability.
2. Speech-to-Text Conversion
Transcribed audio using Whisper AI, dynamically switching between: Base model for English content and Medium model for improved Arabic accuracy.
3. Language Detection
Automatically detected spoken language in each video to decide whether translation was needed.
4. Translation (if Arabic)
Arabic transcripts were translated into English using high-quality NLP translation to maintain compatibility with the English-trained model.
5. Text Cleaning
Removed: URLs, Punctuation & special characters, Irregular spacing, Uppercase casing (converted to lowercase)
6. Tokenization
Split text into tokens (individual words) to standardize formatting and prepare for vectorization.
7. Text Vectorization
Applied a custom-trained FastText model to convert tokens into 300-dimensional embeddings. Combined title, description, and transcript embeddings into a single 900-dimension input vector.



Machine Learning Model

We developed a deep learning model that classifies YouTube videos as Safe or LGBTQ-related based on their title, description, and transcript. The architecture combines CNN to detect semantic patterns and LSTM to capture word context and sequence. Input features were vectorized using FastText, resulting in a 900-dimensional vector per video.

The dataset included over 14,000 manually labeled videos. Transcripts were generated using Whisper AI, then cleaned, tokenized, and embedded before training. The model was trained on Google Colab with GPU support.

To enhance performance, we added a reinforcement learning phase using a custom reward function that encourages confident and correct predictions. This helped the model adapt and reduce false positives.

Total number of samples used for evaluation: 2580		
Final Evaluation Results Table (With Confidence)		
Metric	Before RL	After RL
Accuracy	95.43	97.29
CHAP Score	95.42	97.29
ROC AUC	95.45	97.28
Precision (Not LGBT)	91.58	98.17
Recall (Not LGBT)	100.00	96.34
Precision (LGBT)	100.00	96.44
Recall (LGBT)	90.90	98.23
Correct Predictions (Not LGBT)	1284.00	1237.00
Wrong Predictions (Not LGBT)	0.00	47.00
Correct Predictions (LGBT)	1178.00	1273.00
Wrong Predictions (LGBT)	118.00	23.00
Average Confidence (Correct Predictions)	45.99	50.87

Conclusion

Fitra provides a safe, AI-driven platform that filters LGBTQ+ content and supports strong parental control. Despite limited resources, we achieved our goals and aim to make Fitra a trusted space for children's digital experiences.

Results

Fitra successfully delivered a safe and intelligent video platform for children, integrating AI-powered filtering with a user-friendly mobile application. The app features child profiles, content alerts, watch and search history, and full parental control. User testing demonstrated smooth performance, accurate filtering, and high usability for both children and parents.

Demo



Tools

